BUSINESS INTELLIGENCE LAB

EXPERIMENT 8

Report Scheduling, Automation and Alerts using Microsoft Power BI

Stepwise Procedure

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1. AIM

To design and implement report scheduling, automation, and alert mechanisms using Microsoft Power BI and develop strategies for automated report delivery, monitoring, and business performance tracking.

2. SOFTWARE / TOOLS REQUIRED

- Microsoft Power BI Desktop
- Microsoft Power BI Service
- Global Superstore Dataset.xlsx
- Microsoft Excel

3. PROCEDURE

Part A – Scheduled Reporting Plan

Step 1: Open Microsoft Power BI Desktop and load the Global Superstore dataset.

Step 2: Open the previously created Global Superstore report containing sales, profit, regional, and category-wise analysis.

Step 3: Identify the reporting requirements: Daily Sales Report, Weekly Profit Report, and Monthly Executive Dashboard.

Step 4: Define the reporting frequency and intended users for each report.

Step 5: Prepare the scheduled reporting plan.

Report	Frequency	Intended Users
Daily Sales Report	Daily	Sales Team and Sales Manager
Weekly Profit Report	Weekly	Regional Managers and Finance Team
Monthly Executive Dashboard	Monthly	Senior Management

The reporting plan was designed as a planning deliverable. A recurring delivery schedule was not activated because the Power BI Service displayed a paid-license upgrade requirement for paid sharing/subscription features.

Part B – Automated Distribution Design

Step 6: Identify the user groups that require business reports.

Step 7: Assign suitable reports to each user group.

Step 8: Select delivery methods such as Email, PDF, Shared Workspace, or Power BI Service.

Step 9: Prepare the report distribution matrix.

User Group	Report	Frequency	Delivery Method
Sales Team	Daily Sales Report	Daily	Email
Regional Managers	Weekly Profit Report	Weekly	Email / PDF
Finance Team	Weekly Profit Report	Weekly	Shared Workspace
Senior Management	Monthly Executive Dashboard	Monthly	Power BI Service

Part C – KPI Alert Configuration

Step 10: Identify the important KPIs to be monitored.

Step 11: Define alert conditions for the selected KPIs.

Step 12: Identify the recipients who should receive notifications when an alert is triggered.

Step 13: Define suitable actions for each alert condition.

KPI	Alert Condition	Recipient	Recommended Action
Profit Margin	Below 10%	Finance Manager	Investigate costs and discounts
Sales	Decline greater than 15%	Sales Manager	Analyze the decline and prepare a recovery plan
Returns	Increase greater than 20%	Operations Manager	Investigate product quality and return reasons

Step 14: Publish the Power BI report to Power BI Service using the Publish option in Power BI Desktop.

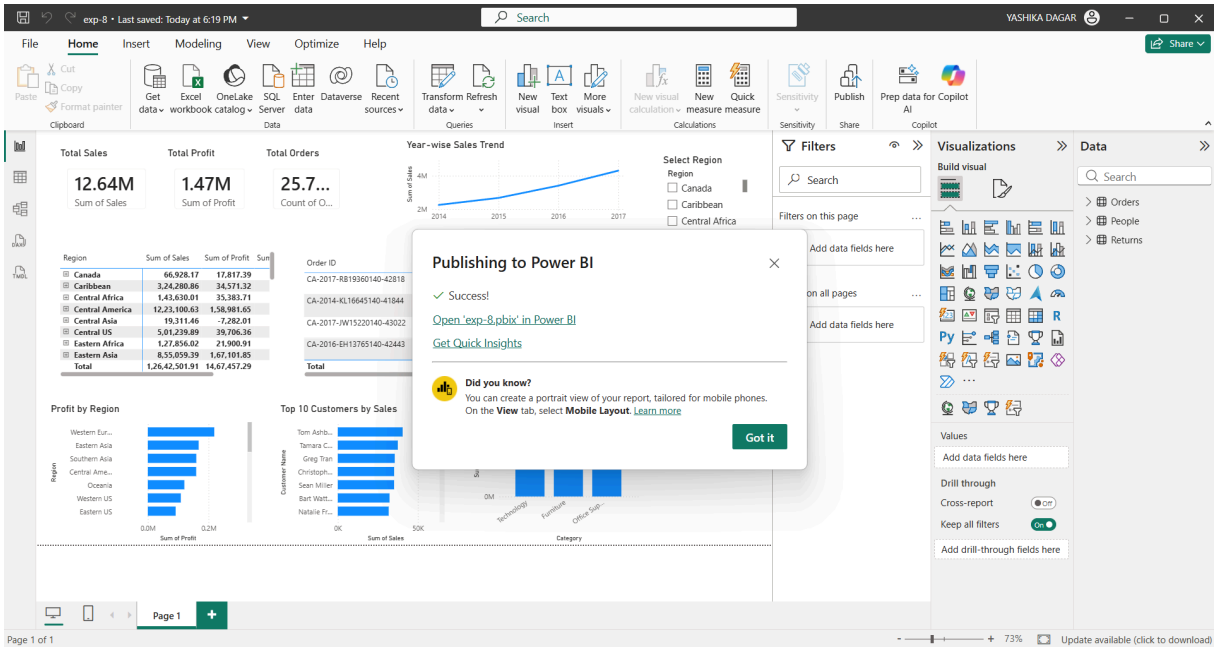


Figure 1: Successful publishing of the exp-8 report to Power BI Service.

Step 15: Open My workspace in Power BI Service and verify that the report and semantic model are available.

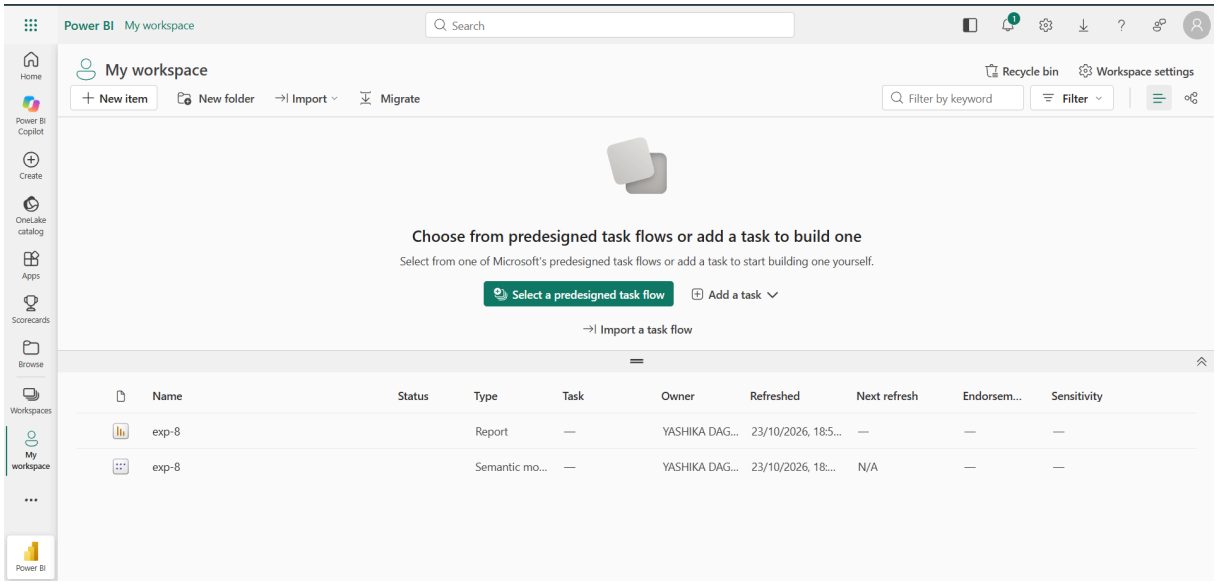


Figure 2: The exp-8 report and semantic model listed in My workspace.

Step 16: Pin supported KPI visuals to a new dashboard named “Experiment 8 KPI Monitoring”.

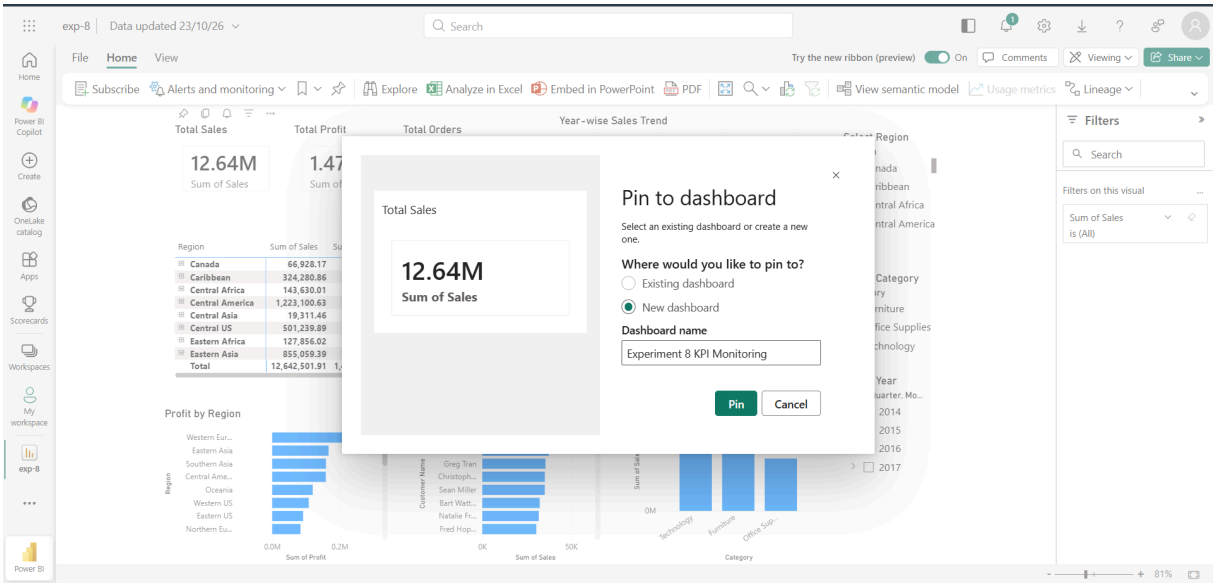


Figure 3: Selecting the new dashboard while pinning the Total Sales KPI tile.

Step 17: Configure a Sales Performance Alert on the Total Sales tile. The saved rule uses an Above condition with a threshold of 12,642,501.91. This is a fixed-value alert and is not an automatic calculation of a 15% decline.

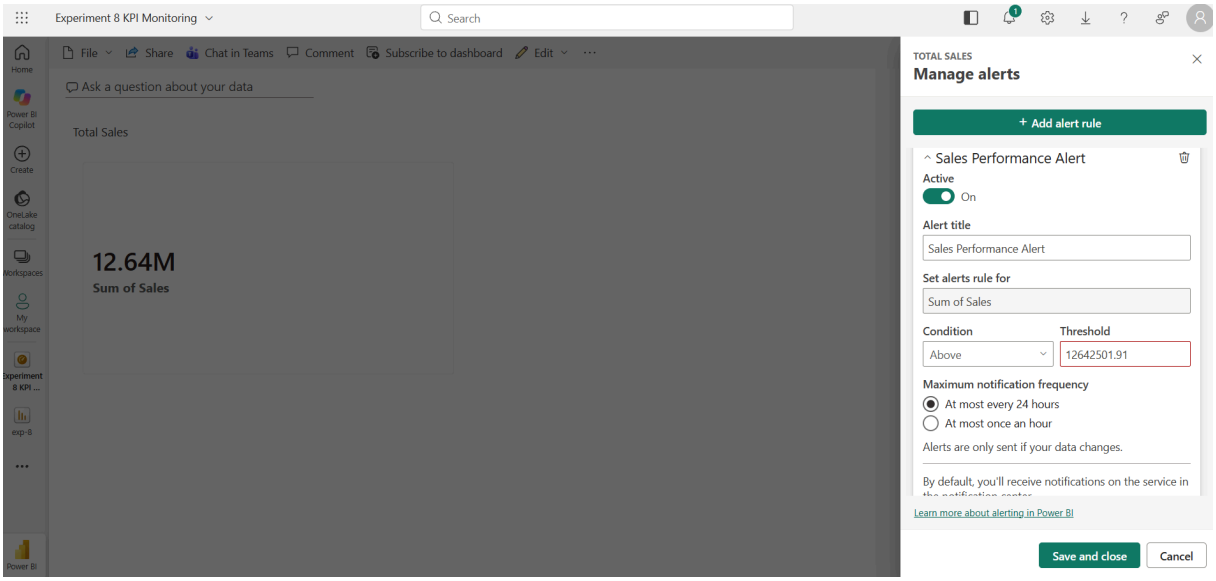


Figure 4: Sales Performance Alert configuration in Power BI Service.

Step 18: Configure a Profit Margin Alert with the condition Below 10.

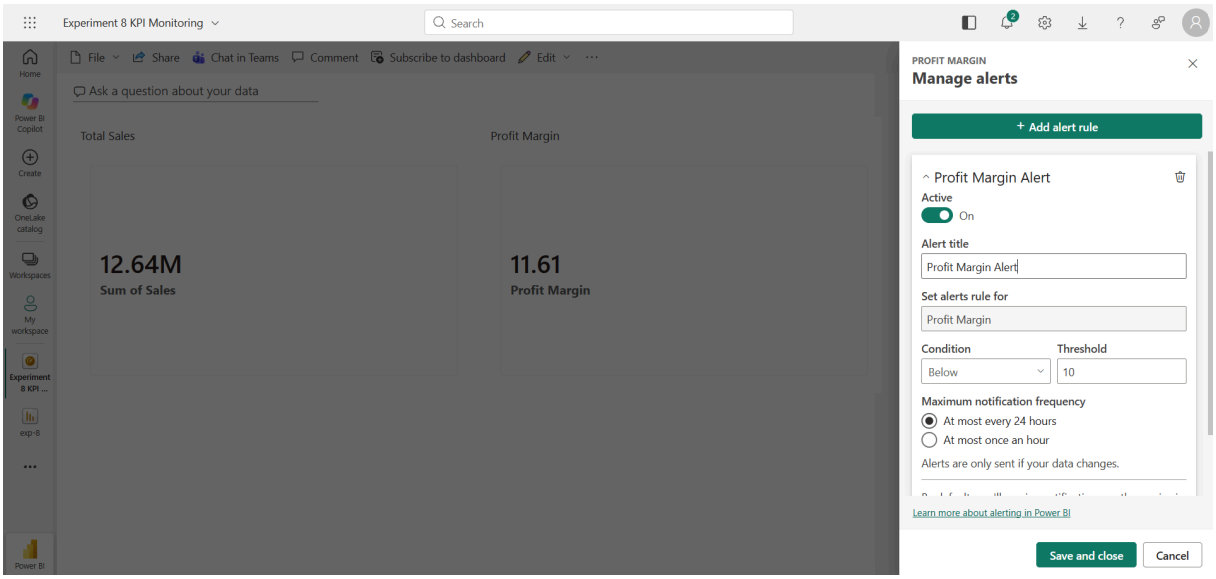


Figure 5: Profit Margin Alert configuration (Below 10).

Step 19: Create and pin a Total Returns KPI card to the dashboard.



Figure 6: Total Returns KPI card created from the Returns data.

Step 20: Configure a Returns Monitoring Alert with the condition Above 2,400. This is a fixed threshold based on the displayed baseline value, not a dynamic 20% increase calculation.

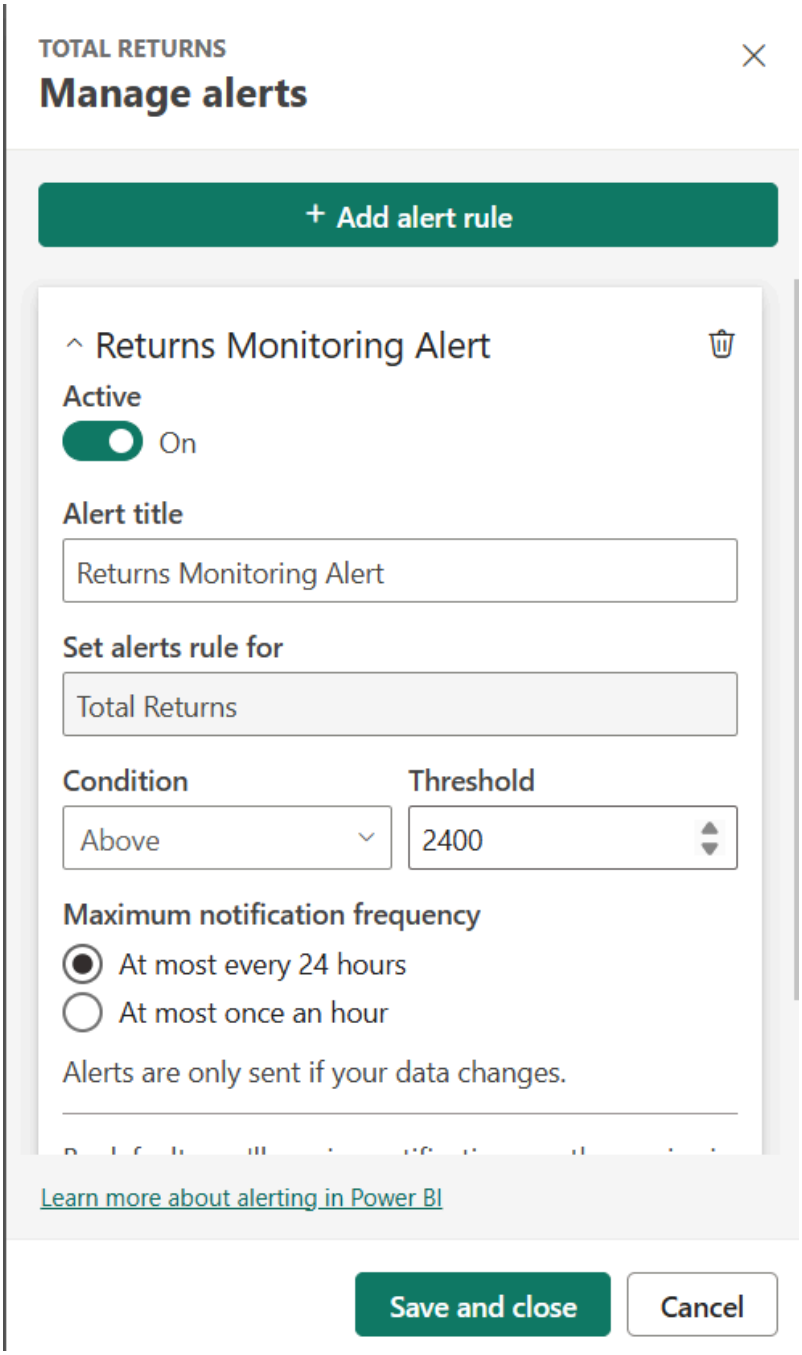


Figure 7: Returns Monitoring Alert configuration (Above 2,400).

Step 21: Review the final KPI monitoring dashboard containing Total Sales, Profit Margin, Total Returns, and Profit by Region.

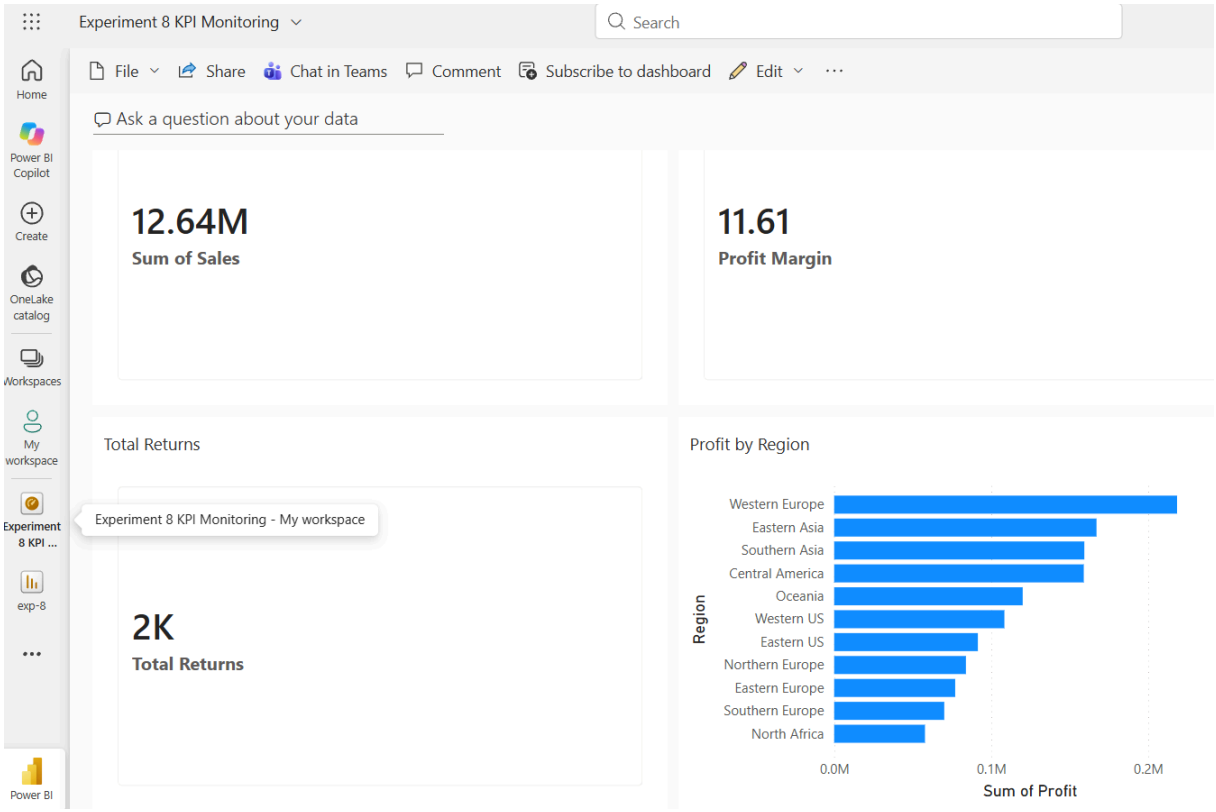


Figure 8: Final Experiment 8 KPI Monitoring dashboard with the regional profit visual.

Note: Power BI data alerts are supported only on eligible dashboard tiles and are based on fixed thresholds. The required relative-change rules (sales decline greater than 15% and returns increase greater than 20%) are therefore documented as monitoring rules; the configured fixed thresholds should not be described as implementing those relative calculations.

Part D – Monitoring Scenario Analysis

Step 22: Consider a scenario in which a regional branch experiences declining profitability.

Step 23: Analyze sales, profit, and profit margin by region using the report.

Step 24: Identify possible alert triggers: Profit Margin below 10% and Sales decline greater than 15%.

Step 25: Identify responsible recipients such as the Regional Manager, Finance Manager, and Sales Manager.

Step 26: Recommend corrective actions: review discount strategies, identify low-performing products, analyze category-wise profitability, reduce unnecessary operating costs, and develop a sales improvement strategy.

Step 27: Prepare a monitoring and recommendation report containing alert triggers, recipients, and corrective actions.

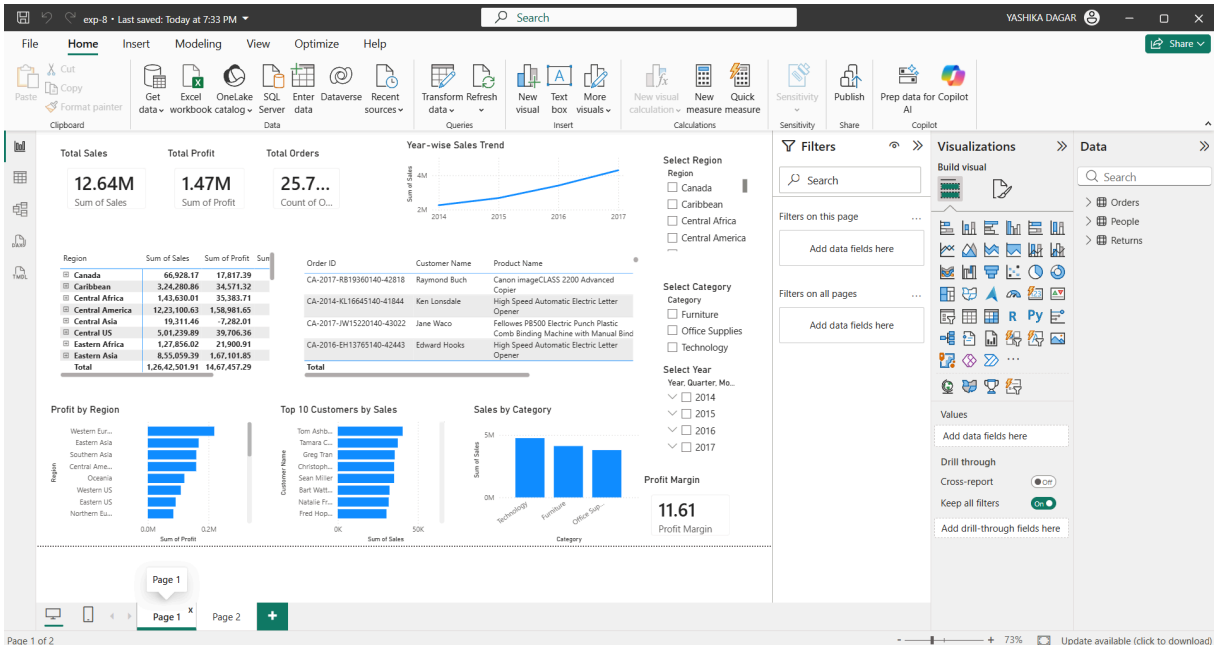


Figure 9: Power BI Desktop report showing sales, profit, category, regional analysis, and profit margin.

4. RESULT

Successfully published the Power BI report to Power BI Service, created an Experiment 8 KPI Monitoring dashboard, pinned Total Sales, Profit Margin, Total Returns, and Profit by Region visuals, and configured fixed-threshold alerts for Sales, Profit Margin, and Returns. The scheduled reporting plan, distribution matrix, and monitoring recommendations were documented. Paid subscription/sharing features were not activated because the service prompted for a paid Power BI license.